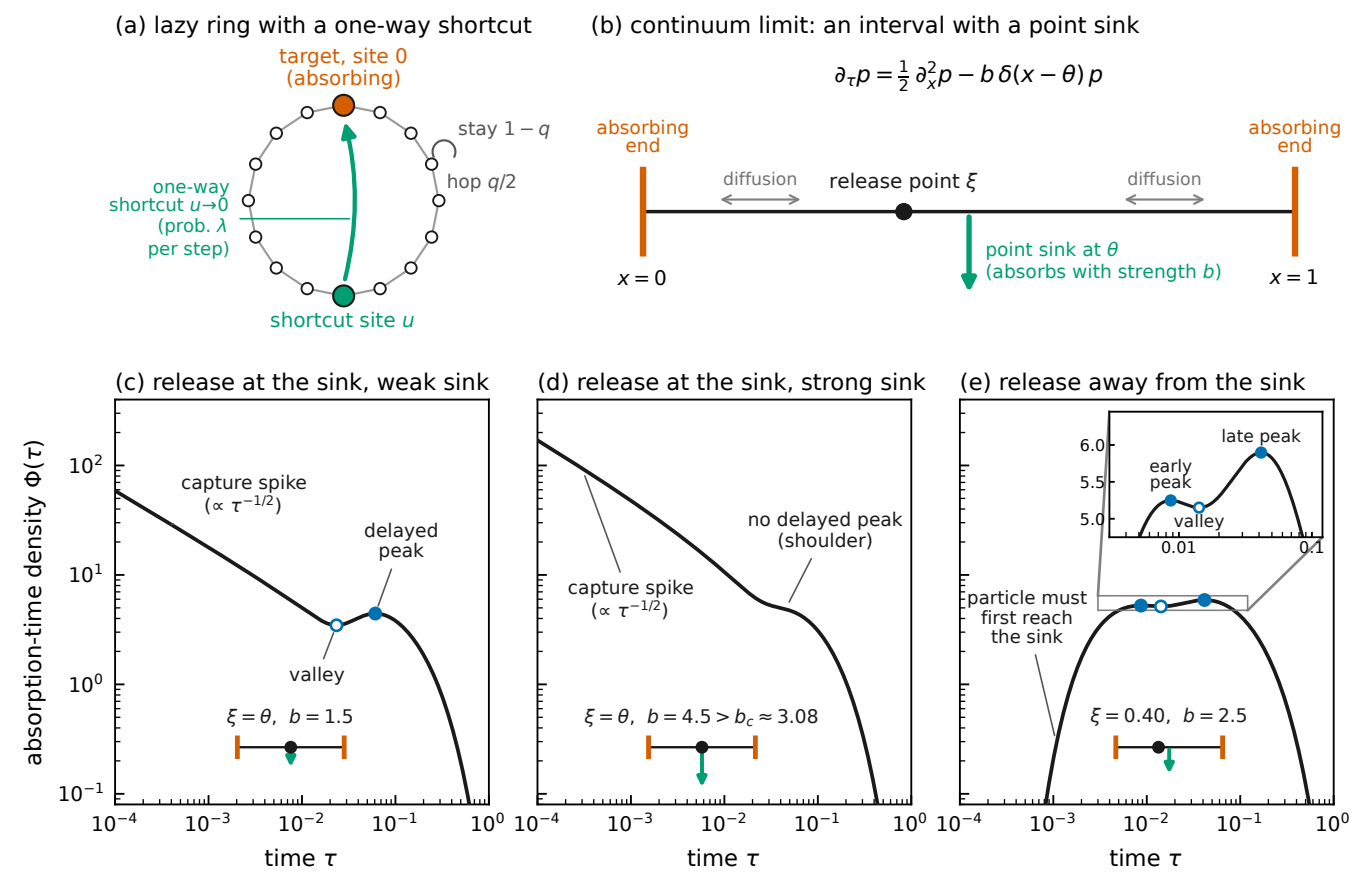
(a) lazy ring with a one-way shortcut
target, site 0
(absorbing)
stay $1-q$
hop $q/2$
one-way
shortcut $u \to 0$
(prob. λ
per step)
shortcut site u
(b) continuum limit: an interval with a point sink
$\partial_\tau p = \frac{1}{2}\partial_x^2 p - b\,\delta(x-\theta)\,p$
absorbing
end
diffusion
release point ξ
diffusion
absorbing
end
point sink at θ
(absorbs with strength b)
$x = 0$
$x = 1$
(c) release at the sink, weak sink
capture spike
($\propto \tau^{-1/2}$)
delayed
peak
valley
$\xi = \theta,\ b = 1.5$
absorption-time density $\Phi(\tau)$
time τ
(d) release at the sink, strong sink
no delayed peak
(shoulder)
capture spike
($\propto \tau^{-1/2}$)
$\xi = \theta,\ b = 4.5 > b_c \approx 3.08$
time τ
(e) release away from the sink
late peak
early
peak
valley
particle must
first reach
the sink
$\xi = 0.40,\ b = 2.5$
time τ